Operating Systems

File and Storage Systems

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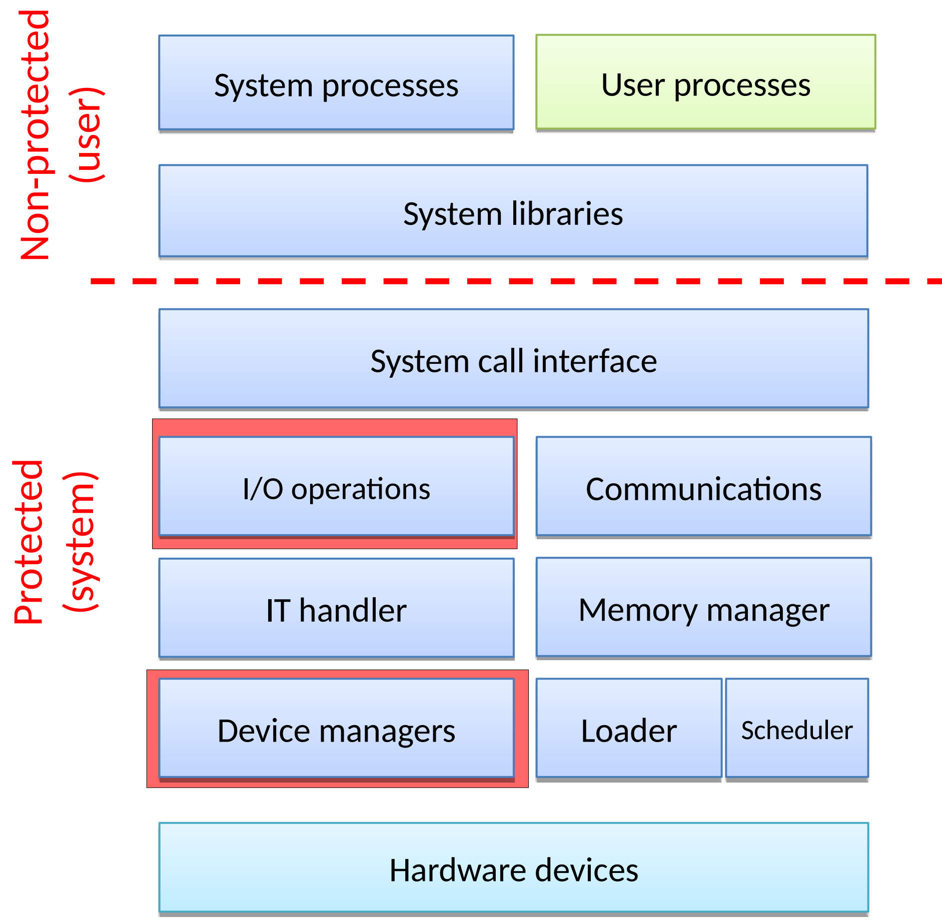
Previously ...

- Tasks
  - typically I/O intensive
  - perform many file operations
  - program code is in the filesystem
- Memory management
  - uses the disk storage to extend the physical memory
- Communication
  - over files (mmap)
- Lab exercises
  - Linux: network filesystem (Samba)
  - Windows: file properties, sharing, network drives

OS architecture

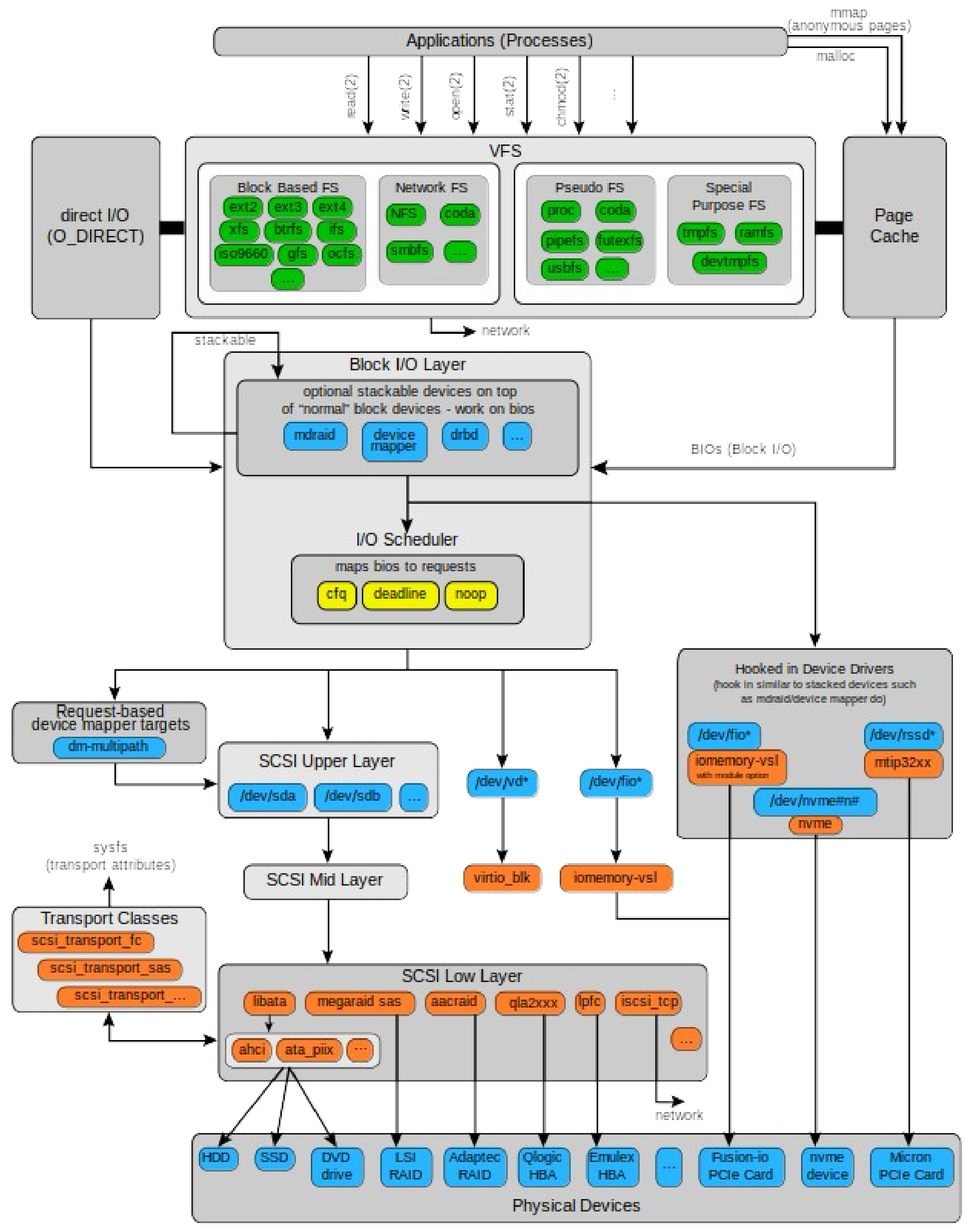


Overview

- The user perspective
  - end user
  - administrator
  - programmer

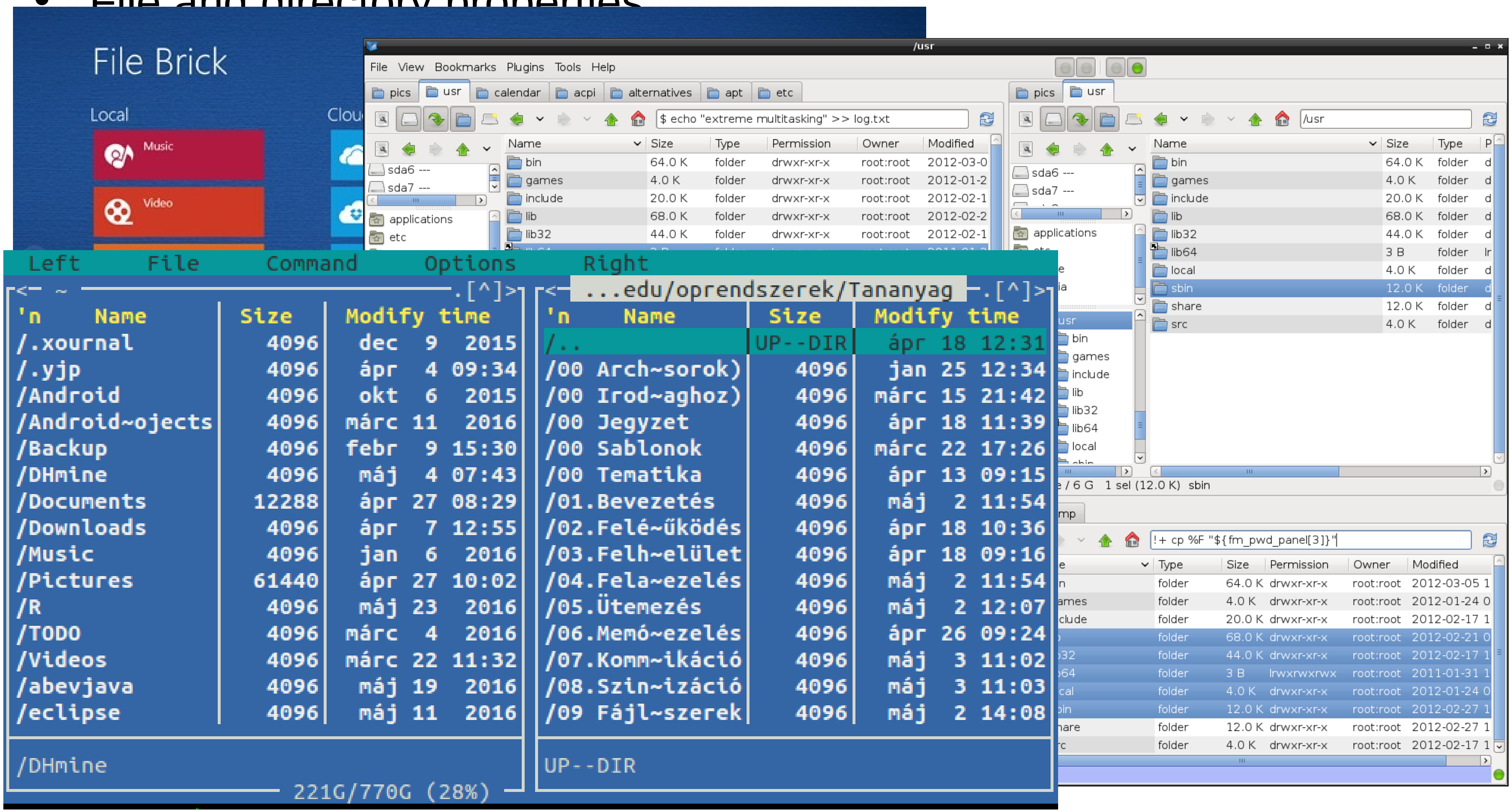
- Internals
  - filesystem APIs
  - disk organization
  - buffer cache
  - virtual filesystems

- Storage
  - devices (HDD, SSD)
  - I/O scheduling
  - virtualization:
    - local (RAID, LVM)
    - network (SAN, NAS)
  - distributed



The end users' perspective

- Command line and GUI tools
- Organizational structure (places)
- File and directory properties



The Admin's and Programmer's perspective

- System Administrator
  - create, check and remove file systems
  - mount local and networked drives
  - performance tuning
  - disk usage
  - backup

- Programmer
  - APIs
    - system calls
    - system libraries
  - file descriptors and operations
    - create, open, read, write, seek, close, delete
  - file locking

Basic concepts

- File
  - logical unit for data storage
  - name (extension), properties
- Directory
  - logical unit for organizing files and directories
  - may contain files and directories
- Volume, drive
  - a set of files and directories
  - typically assigned to a partition on a disk

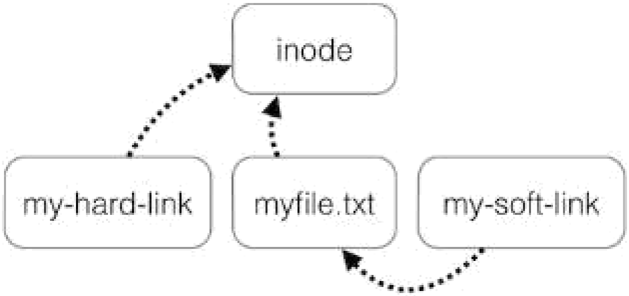
Logical

Physical

- File system
  - stores a coherent set of files and directories
- Partition

Directory structures, volumes and drives

- The basic structure is a directed tree
  - A directory can contain files and other directories
  - The direction of the edges is determined by the containment relation
  - Path: a place of a file or a directory in the tree
    - Absolute: the path from the root of the tree
    - Relative: the path from a specific node in the tree
      - Usually the actual working directory of the user



- Some systems (e.g. Unix) use further edges
  - Hard link
    - linked to the same data
  - Symbolic link (symlink, soft link, shortcut)
    - references a file or directory entry
  - How can we delete the link or data?
  - What happens if there is directed circle in the graph?

- There might be more than one trees in a system
  - There can be more volumes in the system, each one contains one tree

Overview of the Windows 10 folder structure

- More than one folder structures (trees)
  - Physical storages are assigned with logical units, drives
    -
- The boot drive (usually C:) is the starting point (C:\)
  - `\Program Files` – installed applications
  - `\Program Files (x86)` – installed applications (32-bit)
  - `\ProgramData` – user independent data of the applications
  - `\Users` – user folders (files, folders, user dependent application data, …)
  - `\Windows` – the OS files and directories
    - »
- Further drives (D:, E:, …)
  - CD/DVD/USB drives
  - Further partitions on the disk
  - Network file systems
    - »
- Versions, trends
  - In the newer Windows systems the physical storages can be assigned to folders also (not just to volumes), but it isn't a widely-used feature

Overview of the UNIX directory structure

- It is organized into a single tree (no drives)
- The root directory is the starting point (/)
  - `/bin` – binary files for the system
  - `/sbin` – similar to /bin, usually programs with root permissions
  - `/dev` – hardware devices
  - `/etc` – system and application configuration files
  - `/home` – user directories and files
  - `/lib` – basic shared system libraries
  - `/mnt` – the mount point of physical partitions
  - `/tmp` – temporary files (for apps. and users)
  - `/usr` – user programs and libraries, documentation, etc.
  - `/var` – dynamic files of the system, logs, databases, …
  - More details: `man hier`
- Disk usage
  - `df, du, xdu, baobab, kdiskstat, filelight`
- File system „standards”, changes
  - Filesystem Hierarchy Standard (FHS) is just a recommendation
  - UsrMove: the /bin, /sbin is moved under /usr (Solaris11, Fedora)

Overview of the Android directory structure

- Similar to the Unix structure with additional directories
  - `/cache` – cache for applications
  - `/data` – user programs and data
  - `/data/app` – applications installed by the user
  - `/data/anr` – app-not-responding: error logs
  - `/data/tombstones` – memory dumps of the terminated apps.
  - `/data/dalvik-cache` – optimized binary files of the apps.
  - `/data/misc` – user configuration files
  - `/data/local` – temporary files
  - `/mnt` or `/storage` – mounted file systems, e.g. SD card
  - `/mnt/asec` – unsecured copies of the apps. running from SD card
  - `/system` – preinstalled apps., system libraries, configuration files

- Remarks
  - File is system access is limited, root user is inaccessible by default.
  - Apps. stored on the SD card are encrypted (`.android_secure`), these files are mounted under the `/mnt/asec` directory while running

File properties (with Unix examples)

- List the content of the actual directory (ls -la)
  - ```
    drwx------  6 root root   4096 Feb 23 14:20 .
    drwxr-xr-x 22 root root   4096 Nov 21  2014 ..
    -rw-r--r--  1 root root    570 Jan 31  2010 .bashrc
    -rw-r--r--  1 vps  vps   71103 Nov  5  2013 package.xml
    -rwxrwxrwx  1 root root     35 Feb 23 14:21 test.sh
    lrwxrwxrwx  1 root root      8 Nov 24  2014 www -> /var/www
    ```
- What is in the list?
  - Type of the entry: - d p l b c s
  - POSIX permissions (see next slide)
  - Number of links
  - Owner and group
  - Size
  - Timestamp (ctime: change of the metadata, mtime: data modification, atime: access time)
  - Name of the entry
- The OS also stores
  - Unique identifier (for internal identification)
  - Location (where the file is stored on the disk)
    ```

The Unix permission systems

- POSIX permissions
  - 3x3 bits: owner, group, others X read, write, execute
  - Values: read-4, write-2, execute-1, no access-0
    - E.g.: 740 = owner: RWX, group: R, others: no access
  - In the case of directories, the execute means „list”
  - Setting: chmod <permissions> <file/directory>
    - E.g.: chmod 750 /home/me   chmod u+rwx,g+rx,o-rwx /home/me
- Special permissions: SETUID, SETGID, StickyBit
  - SETUID/GID: set user ID upon execution" and "set group ID upon execution
    - The executed file will have the same permission as the owner (not the user which executed the file)
    - It is usually set to files which require root permissions
  - StickyBit: only the owner (and root) can delete/rename the files or directories
  - `drwxrwxrwt 44 root root     12288 máj    9 15:25 /var/tmp`
- POSIX ACL (access control list) (*extended permissions*)
  - flexible, can store several access control lists for an entry

Sysadm tasks

- Create (format)
  - type (next slide)
  - properties
  - name (for humans), ID (for machines)
  - storage (disk, network etc.)
- Mount
  - physical → logical assignment
  - mount point
  - mounted file system covers a part of the file system tree
- Check, tune
  - offline checking
  - change size of storage has changed
  - tune for performance, compression etc.
- Backup (see later)

An overview of the widely used file systems

- FAT32
  - Typically used on portable storage devices because the compatibility
  - Originally 8+3 character file names extended to 255 characters, maximum file size: 4GiB (!)
- NTFS
  - Default file system in Windows
- UFS/ Berkeley FFS
  - Traditional UNIX file system, currently rarely used
- ext2,3,4 (cased on UFS)
  - Currently used file systems in Linux systems
- XFS
  - Default in RedHat Linux 7
- HFS+
  - Default in MacOS
- Integrated file + virtual storage systems (see later)
  - ZFS: Designed for Solaris, later it become open source, popular in BSD-s also
  - Linux btrfs: newer, currently under development
- Many more file systems

Practice in Linux

- Basic file and directory operations
  - `cp, mv, cd, pwd, mkdir`
    - How to rename a file?
- File attributes: `ls -la`
- Managing file systems: `mount, umount, df, mkfs, fsck`
- Example: create a file system in a file
  - ```
    dd if=/dev/zero of=filesystem.img bs=1k count=1000
    losetup /dev/loop0 filesystem.img
    mke2fs /dev/loop0
    mount /dev/loop0 /mnt
    ```
  - An annoying error: device is busy
    - while unmounting a file system
    - check what is used: `lsof /mnt`
- What's happening in the file system?
  - `iotop, sar, dstat, vmstat, …`
  - `sudo sysctl vm.block_dump=1`
  - `tail -f /var/log/kern.log`
    ```

Backing up and restoring data

- Multiple causes of data loss
  - Uncorrectable fault in the file systems
    - The error in the physical storage (disk error)
    - Inconsistency caused by power failure or other HW error
  - User mistakes (not rare)
    - Accidental deleting of files or whole file systems, partitions
  - Malwares (sadly these are also not rare)
    - Deleting or encrypting data (ransomware)
- The type of data loss
  - Limited (e.g.: disk error, user mistakes, …)
  - Total (e.g.: SSD sudden death)
- Creating a backup
  - How: automated (regular), manual (casual)
  - What: files or whole file system
    - A consistent state has to be backed up – problematic when the FS is in use
  - Where: high capacity disks, CD/DVD, tape systems
- Restoring the system from a backup (recovery)
  - Bare metal recovery: restoring the whole system
  - Data recovery: only recovering specific files

The programmer's perspective

Programming interfaces

- Opening (and creating) files
  - `open()` system call and its arguments
  - File descriptor and the opened file object (metadata)
  - File opened by multiple processes?
- Read, write, seek: `read()`, `write()`, `fseek()`
  - Sequential access: the data is accessed in the stored order
  - Direct access: given sized blocks can be read in any order
- Close files: `close()`
- Managing directories:
  - `opendir()`, `readdir()`, `rewinddir()`, `closedir()`

Locking files

- Locking files
  - a file may be a shared resource
  - synchronization problem: keep the file content consistent
  - we may use any synchronization method
  - the kernel also provides efficient and better locking methods for files
  - note: deadlocks are also possible
- Advisory locking
  - the OS provides tools for the tasks but they may ignore these
  - tools used → locking works
  - tools not used → no locking
- Mandatory locking
  - locking enforced by the kernel
- The scope of locking
  - whole or a part, see Windows LockFileEx(), Unix fcntl()

Shared access to files through memory (mmap)

- Communicate through a file
  - It is problematic with the standard op.-s (read(), write(), fseek())
  - Can we use a file like the shared memory?
- UNIX mmap, Windows: CreateFileMapping
  - An open file object (open()) can assigned to an address: mmap(addr, size, prot, flags, fd, offset)
    - addr: the assigned address, 0: the kernel choses
    - size: the accessed data range
    - prot: the mode of access: R, W, X
    - flags: own or shared file, etc.
    - fd: file descriptor returned by the open() system call
    - offset: the start position
  - Return value: the assigned virtual memory address
  - Close the assignment: munmap(addr, len)

I/O operations without waiting

- Non blocking I/O
  - syscall options not to block the task
  - the call returns immediately
    - with the data
    - or with „no data” error
  - the task may/should retry the operation any time

- Asynchronous I/O
  - the task initiates the I/O and sets a buffer for the data
  - the asynchronous I/O is performed in the background
    - the system call returns immediately
  - the task can perform other instructions
  - when the I/O is done, the kernel notifies the caller
    - e.g.: with a signal with custom handler
  - see POSIX aio, Windows I/O Completion ports

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  - virtual file systems
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  - I/O scheduling
  - virtualization:
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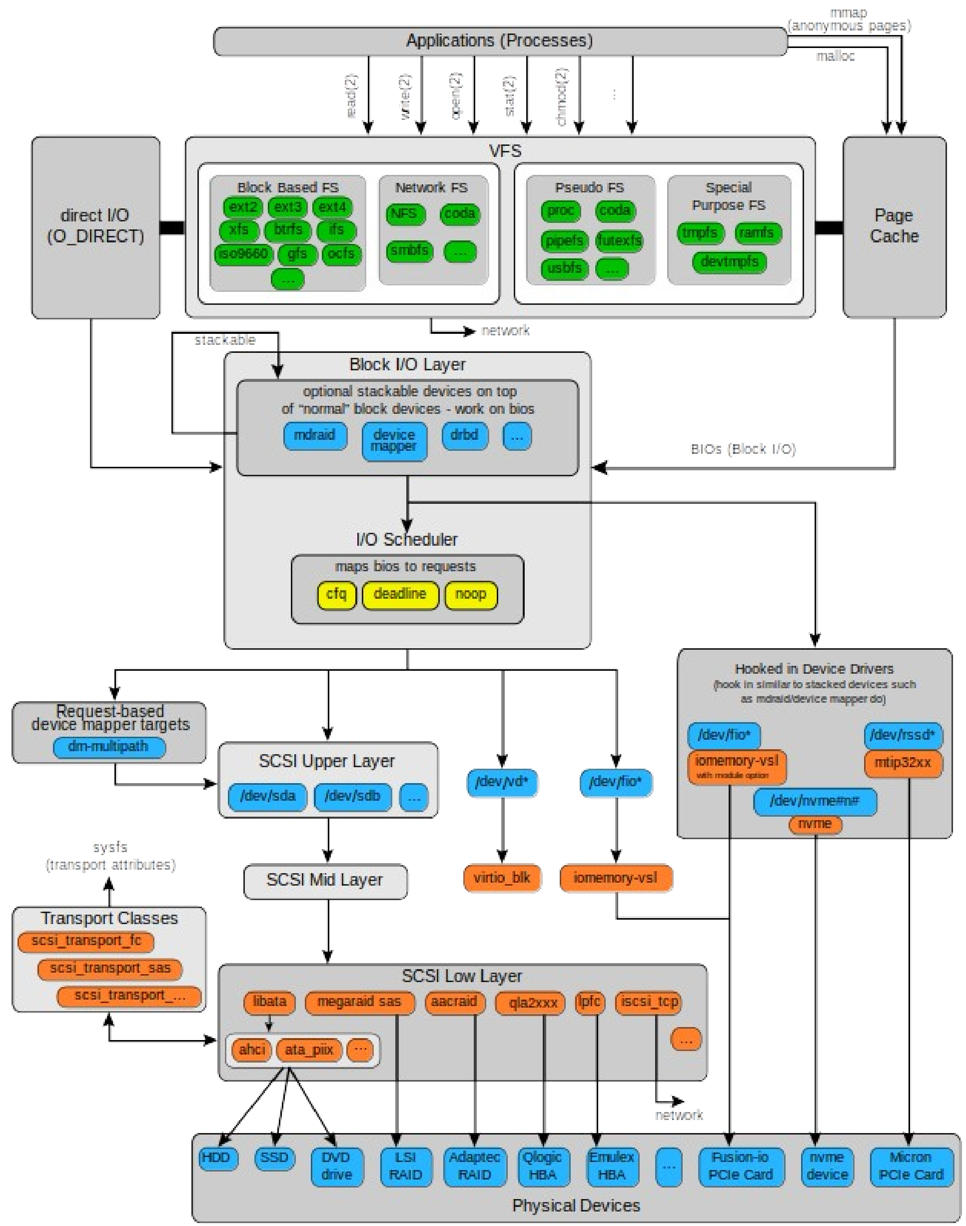


Implementation of file systems (overview)

- Operation from the user’s point of view (already discussed)
  - Files, directories, tree/graph structure
  - Format, mount, unmount
  - Check, repair, create, modify, tune
- Organizing the file system in the storage
  - The logical units are assigned to physical devices
  - The data is stored in blocks
  - Beside the file contents, metadata is also stored
  - Managing the free (unused) blocks in the storage device
- Run-time operations
  - File system descriptors (metadata of the mounted file systems)
  - Descriptors (metadata) of the files
    - Access to opened files
  - Managing the data in the memory, buffering

File system structure

- The stored data
  - File system metadata (superblock, master file table, partition control block)
  - File metadata (inode, file control block, on Windows: it is part of master file table)
  - Stored data

superblock	file metadata	data blocks

- The file system metadata
  - **On disk**
    - Type and size
    - List of free blocks
    - The location of the file metadata
    - State
    - Modification information
    - …

In memory

- everything that is on the disk
- mount information
- „dirty” bit
- locking info
- ...

- The file system is sensitive to metadata loss (e.g. hardware error)
  - Therefore backups are made
  - See: `dumpe2fs /dev/sda1 | grep -i superblock`

File metadata

- **On disk**
  - Authentication information (UID, GID)
  - Type
  - Permissions
  - Timestamps
  - Size
  - Data block locations
  - Example: UNIX inode (index node), Windows Master File Table entry
- **Runtime extensions** (in memory)
  - State (locked, modified, etc.)
  - Disk/file system identifier
  - Reference counter (file descriptors)
  - Mounting point descriptor

Storing data blocks (allocation methods)

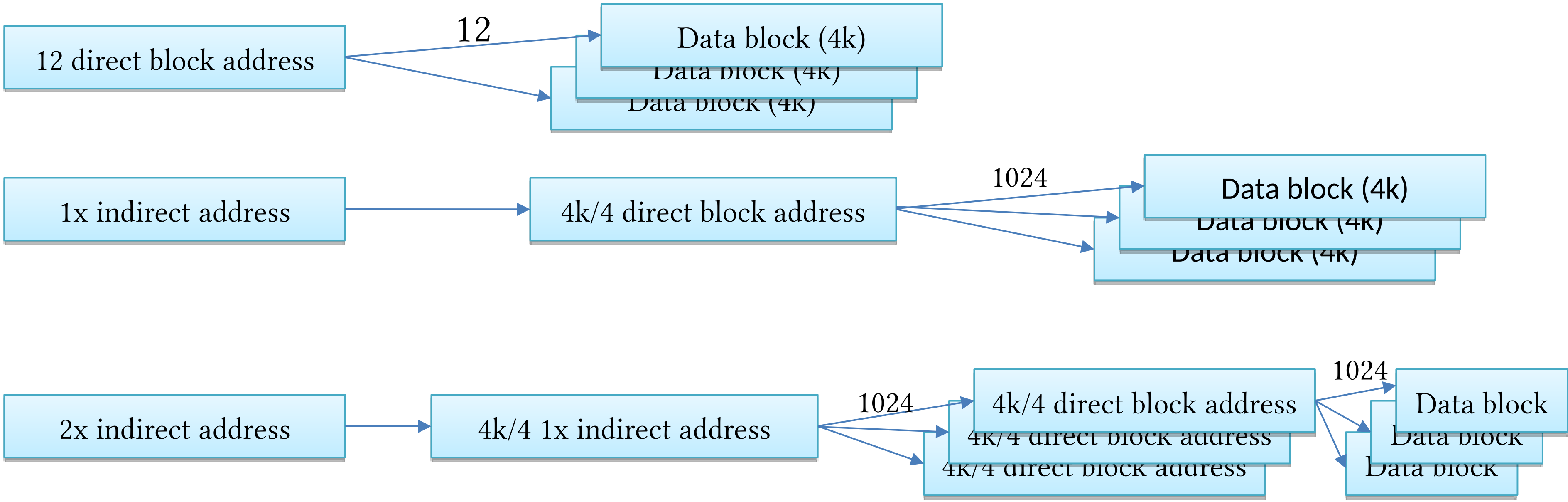
- Continuously on the disk…
  - simple but causes serious fragmentation over time

- Chained list (sequential access), e.g. FAT
  - data is divided into in smaller chunks (blocks)
  - data blocks are stored in a linked list
    - the address of the first part is in the metadata
    - every part contains the address of the next part
  - efficient for sequential access, not good for random access
  - very sensitive to errors

- Indexed storage (direct access)
  - data is divided into in smaller chunks (blocks)
  - the location/map of the blocks: **the index** (see next slide)
  - block may be stored sequentially if possible
  - efficient, good for random and also for sequential access
  - sensitive to loosing the index (may be duplicated)

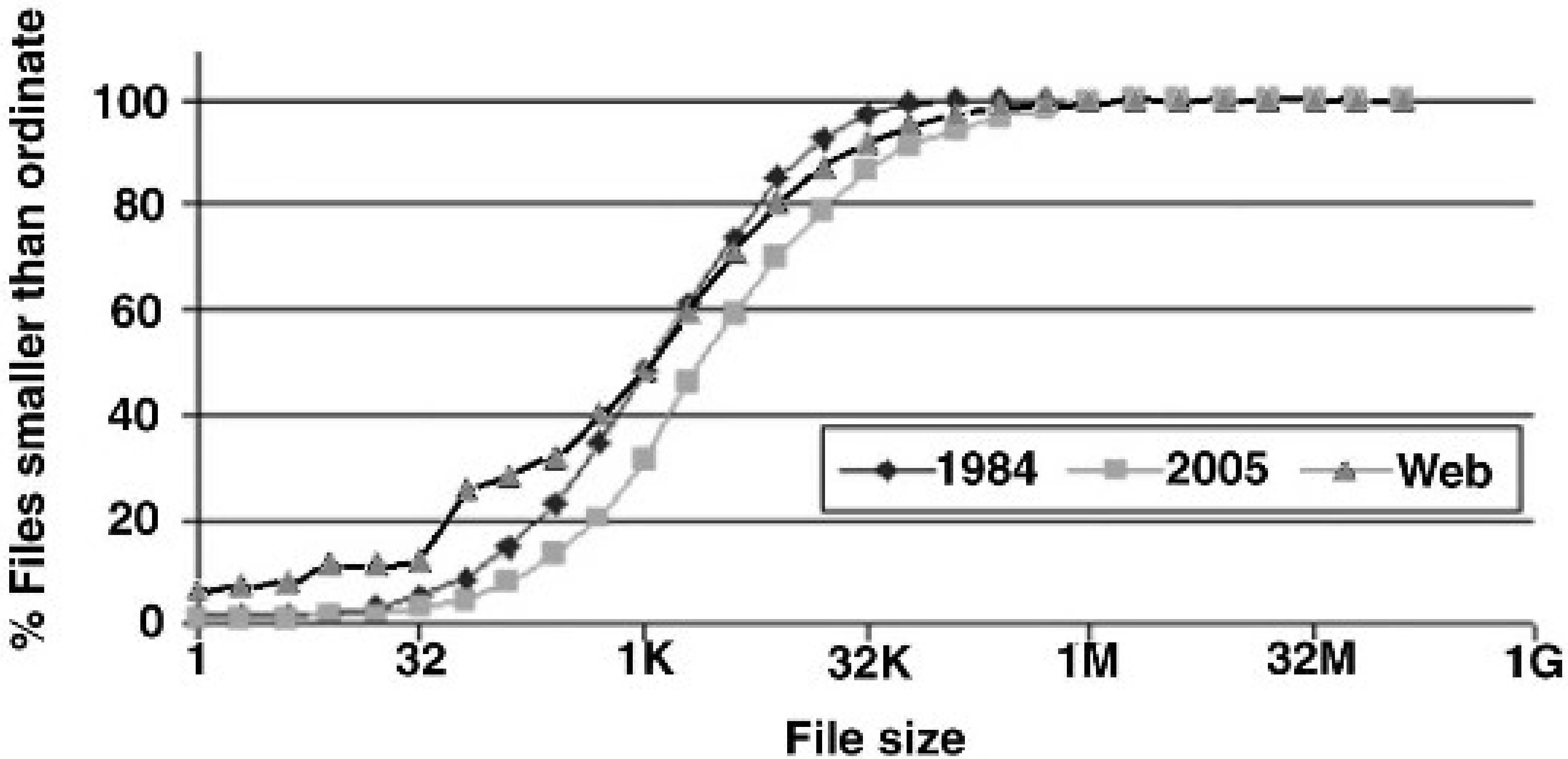
Example: Multiple indexed data block address table

- Index table
  - 12 direct block address
  - single and double indirect block address (to cover large files)
  - 4 kB block size
  - 4 byte address



What is the maximal file size?

How to determine the block size?



Source: Andrew S. Tanenbaum, Jorrit N. Herder, Herbert Bos
File size distribution on UNIX systems: then and now. Operating Systems Review 40(1): 100-104 (2006)

Managing the free blocks

- Bitmap, bit-vector description
  - Every block is represented by a bit
  - 1=free, 0=used
  - Simple method, easy to find a free block
    - The map can be stored in the memory for smaller FS
    - Typically there is a CPU instruction for getting the first non zero bit location
  - It uses more memory for a larger file system
- Chained list storage
  - The free blocks are marked and the address of the next free block is written there
  - Only the address of the first free block has to be stored
  - Simple, but not so efficient method
  - It can be combined with the chained list block allocation method
- Hierarchical methods
  - Managing the group of (free) blocks
  - The groups can be created based on the size of the FS
  - Within a group, a simpler structure can be used (e.g.: bitmap)

Accelerating data transfers

- Disk buffering
  - to accelerate the access to frequently used data
  - works in coordination with virtual memory management (paging)
    - data is loaded into frames by the VMM
    - page replacement may also free disk buffer frames if needed

- Accelerating..
  - read operations
    - read ahead automatically
    - may be instructed using system calls (see posix_fadvise)
  - write operations (when to write the modified data to the disk)
    - **Write through cache**
      - write immediately to the storage
      - slow but reliable
    - **Buffered write**
      - writes data periodically (flush, sync)
      - significantly faster but may cause data loss

Metadata consistency and journaling file systems

- When to write metadata in the memory to the disk
  - similar to data buffering but more complex problem due to transactions
  - a write through cache may cause significant performance loss
    - e.g. file access times are changing rapidly
- Journaling file systems
  - changes are saved to a journal, which is always stored on the disk
    - file system operations are grouped into transactions
    - transaction is finished when it is stored in the journal
    - the transaction data is deleted when it is committed to the filesystem
  - the journal is sequential access circular buffer
  - What happens if the system crashes? The journal is processed during reboots.
- Log-structured file system: the log is the file system (e.g. BSD LFS)
- Copy-on-write file system (ZFS, btrfs)
  - a different solution that works on duplicated metadata structures

Virtual File Systems

- There are many types of file systems
  - Typically under UNIX systems, multiple types used at the same time
  - We can’t except that the programmers manage them separately

- VFS is an implementation independent file system abstraction
  - The basis of the modern Unix file systems

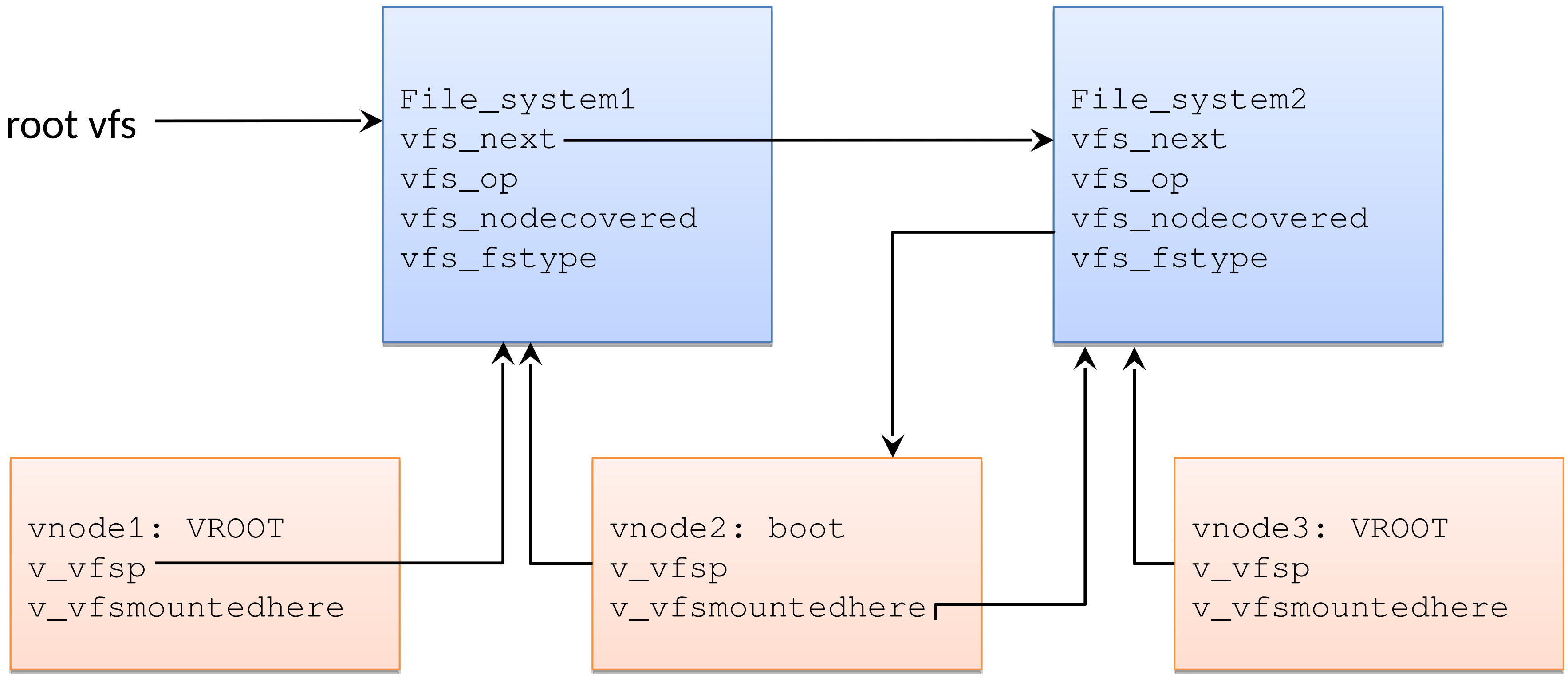
- Goals
  - Supporting multi type file systems running simultaneously
  - Standard programming interface (after mounting)
  - Provide the same interface also for special FS (e.g. network)
  - Modular structure

- Abstraction
  - fs (file system metadata) → vfs
  - inode (file metadata) → vnode

vnode and vfs

- vnode data fields
  - Common data (type, mounting, link counter)
  - v_data: file system dependent data (inode)
  - v_op: table of the file methods (operations)
- vfs data fields
  - Common data (FS type, mounting, vfs_next)
  - vfs_data: file system dependent data
  - vfs_op: table of the FS methods (operations)
- Virtual functions
  - vnode: vop_open(), vop_read(), …
  - vfs: vfs_mount, vfs_umount, vfs_sync, …
  - These are translated to the FS dependent methods

The connection between vfs and vnode



Special virtual file systems (examples)

- Which file systems are supported?
  - `cat /proc/filesystems`
- devtmpfs and devfs
  - accessing the HW devices trough the file system
- procfs
  - accessing to the process metadata and kernel structure through the FS
- sysfs
  - accessing to kernel subsystems through FS
- cgroup, cpuset
  - setting resource allocation for process groups
  - `mount | egrep “cgroup|cpuset”`

Implement your own file system using VFS

- Is not that hard...
- See
  - Ravi Kiran, „Writing a Simple File System”
  - Steve French, „Linux Filesystems 45 minutes” ODP PDF
    - „A Step by Step Introduction to Writing (or Understanding) a Linux Filesystem”

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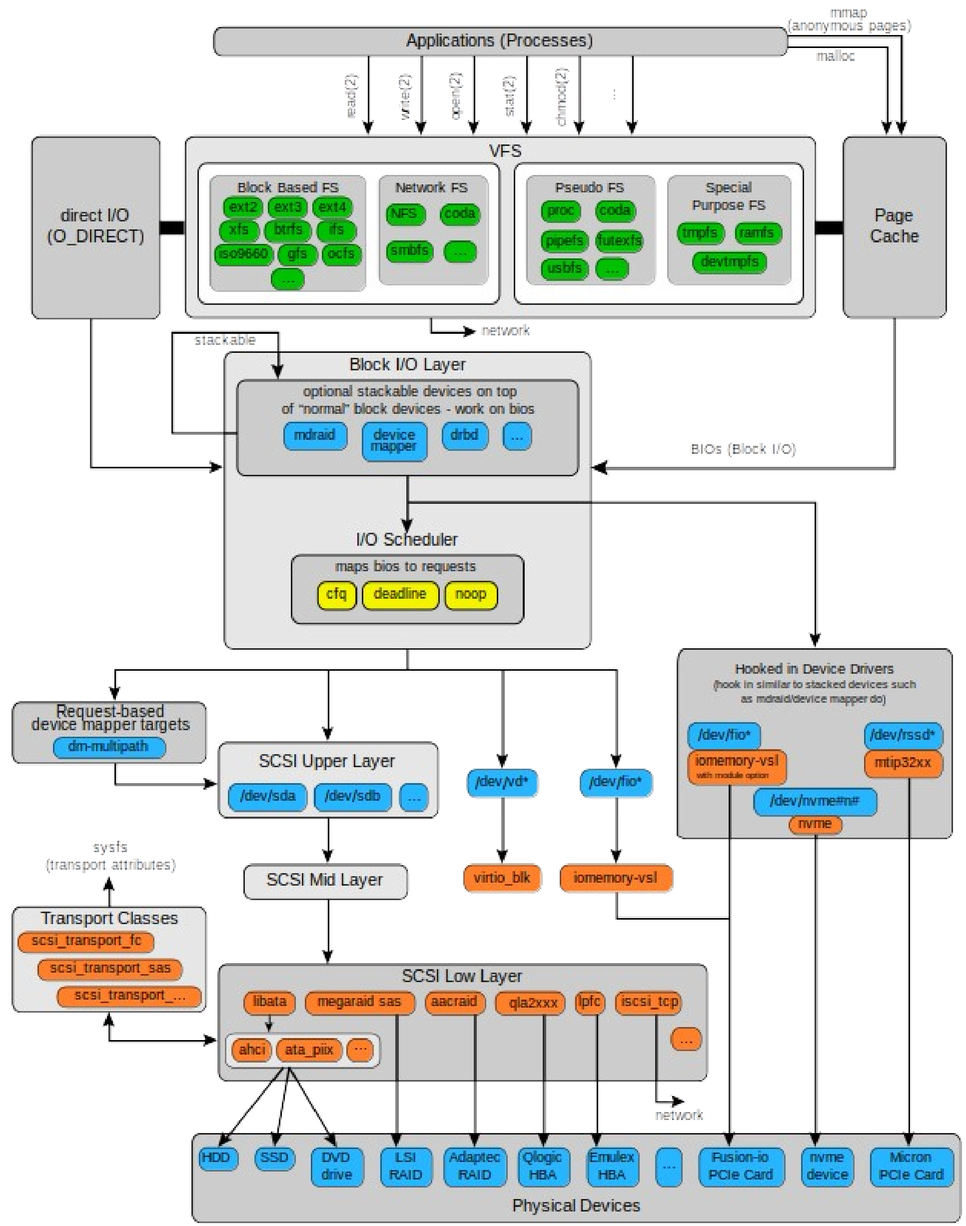
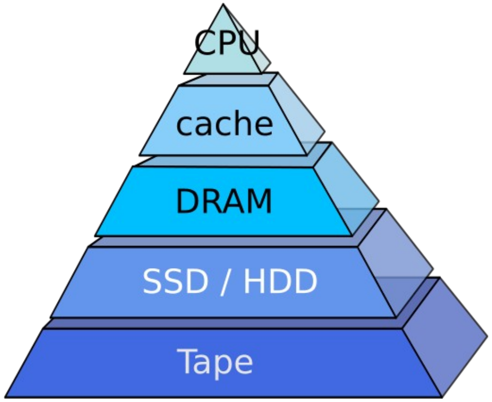


Storage solutions

- Physical devices
  - Magnetic
    - HDD and tape devices
  - Optical
    - CD, DVD, Blu-ray
  - Nonvolatile memories (solid state, integrated circuit based)
    - SSD, USB drive, SD card
- Virtual storage systems
  - extend (capacity, services) other storage systems
    - merging
      - e.g. RAID, LVM
    - network access
      - file or block level transfer
      - e.g. NAS, SAN
    - distributed system
      - For reliable and scalable storage systems
      - e.g. Ceph, GlusterFS
  - In certain cases these are integrated with the FS
    - e.g. Solaris ZFS, Linux BTRFS, …

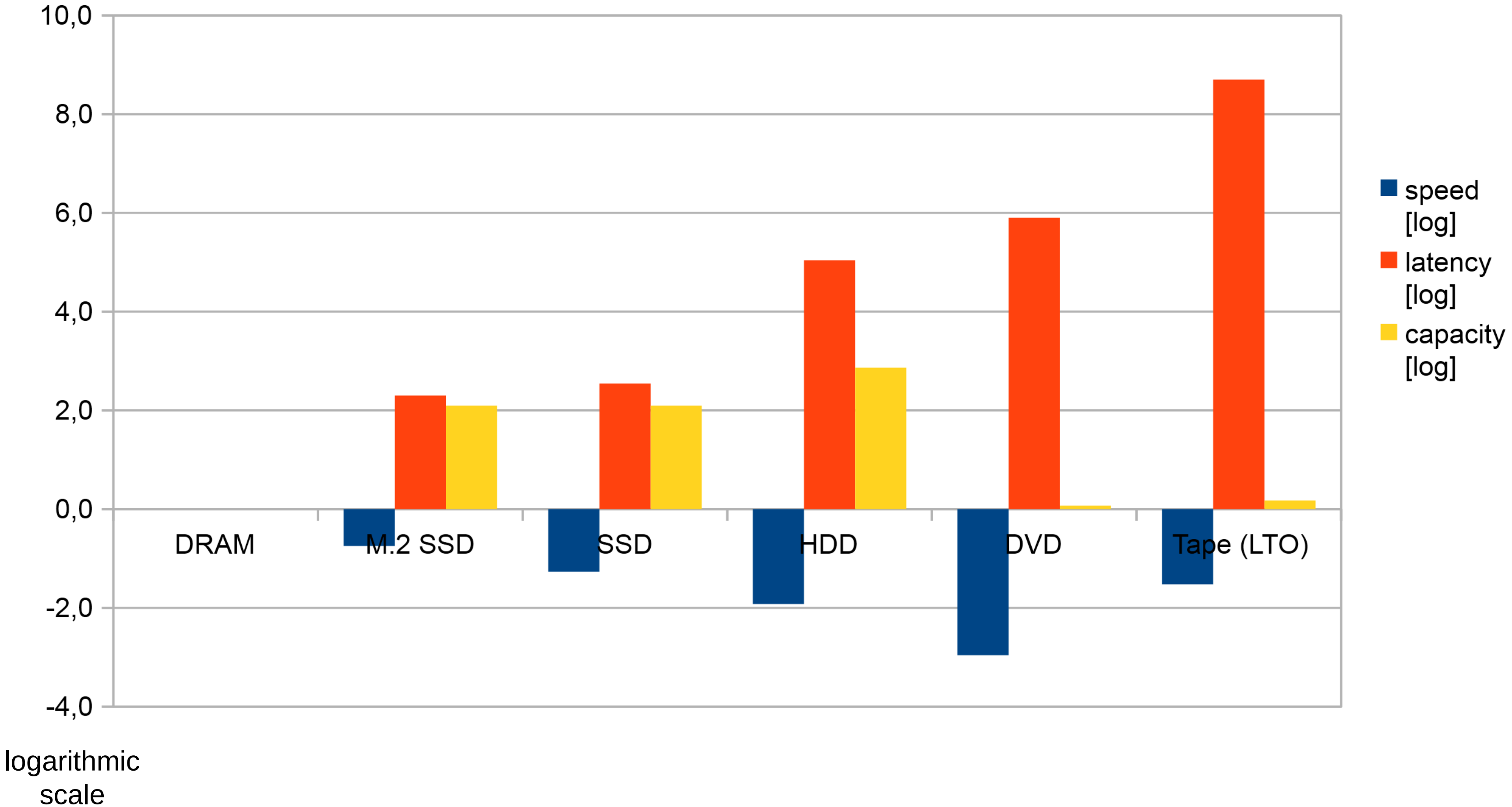
Properties of physical storage systems

- Characteristics
  - capacity from bytes to petabytes
  - throughput (read/write) 10 MiB/s ... 200 GiB/s
  - access time: 0.5 ns ... seconds/minutes



- Reliability
  - measures related to the life-time of a device (see SMART)
  - **Annualized failure rate (AFR)**
    - How many devices fail within a year?
    - Typically 2-4%, but sometimes above 10%
  - **Mean time to failure (MTTF)**
    - Millions of operating hours (>100 years), according to vendors
    - It is related to all of the devices averaged, not for a single device
    - Bathtub curve: higher failure chance for old and new devices
    - Disk failures in the real world: What does an MTTF of 1,000,000 hours mean to you?
  - **Total bytes written (TBW, for memory based devices)**
    - The memory pages cannot written infinite times

Performance compared to DRAM



Trends of physical storage systems

- In the past
  - significant performance difference between CPU and disks
    - The CPUs were developed faster than HDDs
    - “slow I/O” was a design principle for operating systems

- Present and near future
  - the size of the physical memory is greatly increased
    - the size of disk cache is higher
  - new methods based on fast CPU-s
    - runtime data compression (ZFS, btrfs)
    - deduplication
      - avoiding the storage of the same data part more than one time
  - memory based „disks”
    - increasing speed with very low latency
    - **storage class memory**: almost DRAM performance with large capacity

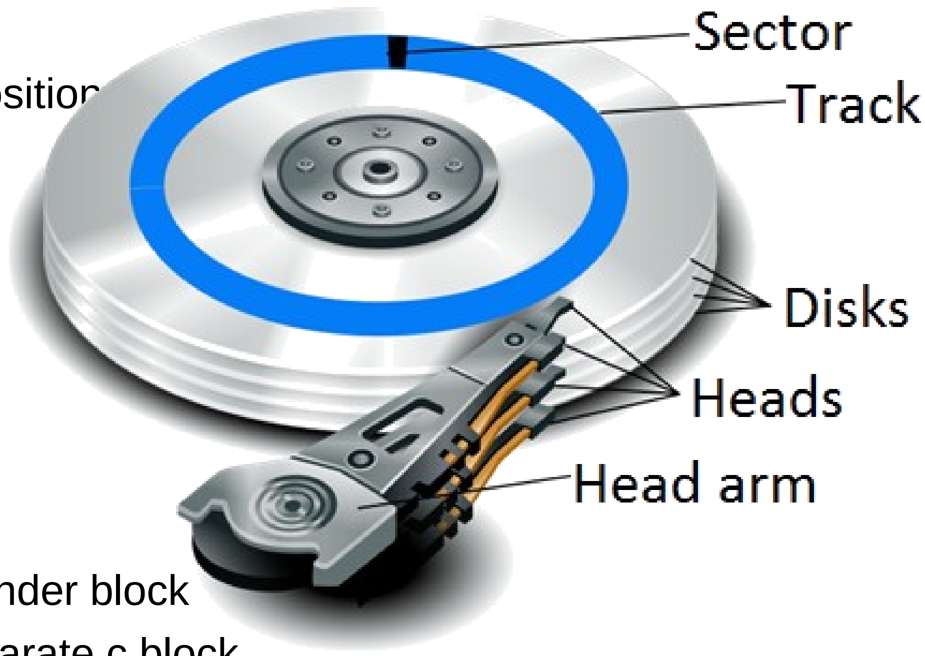
Tape drives

- Traditional tool for backups
  - High capacity
  - Long lifetime
  - slow operation
  - manual / robotized cassette change

- Recent developments
  - high sequential read speed
    - Tape – 300 MB/s, SSD – 500 MB/s
  - larger caches
    - Almost every data is there
    - Filled with sequential read
  - see log-structured file systems
    - sequential read/write
  - good as a general storage?

Magnetic (mechanical) disk drives

- The location of the superblock, inode list, data blocks on the disk
  - Goals: performance, reliability
- Cylinder block
  - Tracks assigned to the same head position
  - The data can be accessed without head movement
  - Collective damage is possible when a head-disk collision happens



- Allocation principles
  - The superblock is stored in every cylinder block
  - inode list and free blocks are in a separate c.block
  - Small files in the same c.block
  - Larger files are distributed between c.blocks
  - The new files will be on a less used c.block

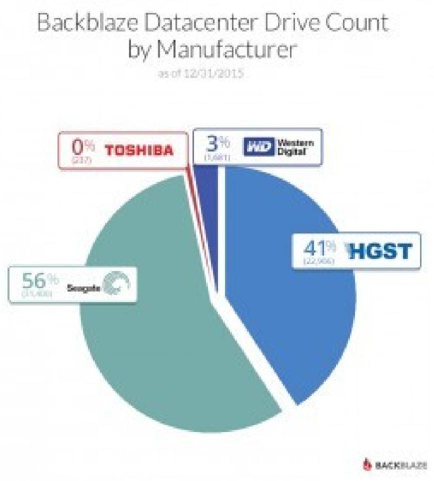
Scheduling of disk operations

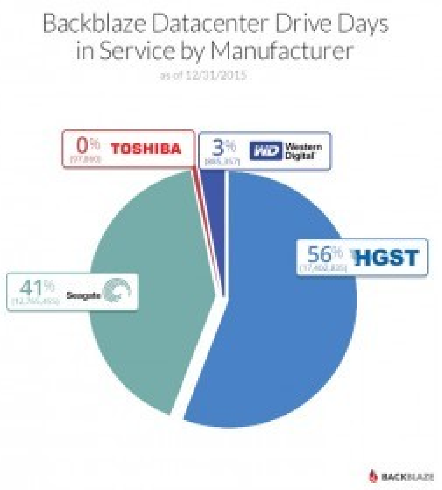
- Scheduling increases the performance
  - especially on mechanical drivers with slow access times

- E.g. Linux I/O Schedulers
  - **Noop**: simple FIFO
    - may concatenate adjacent requests
    - small overhead
    - recommended if the storage system has internal scheduling (RAID, NCQ, VMs)
    - best for CPU intensive systems (low load on disks)
  - **Deadline**: tries to perform requests before a deadline
    - requests are ordered by the block address in read and write batches
    - recommended for I/O intensive systems with many parallel requests
  - **CFQ** (Completely Fair Queuing): equal service for every request
    - has queues for every process and assigns a time-slice to them
    - estimates the load for each queue
    - scheduling depends on this estimation and the priority of the queues
    - recommended for general usage (usually this is the default)

Reliability of hard disk drives

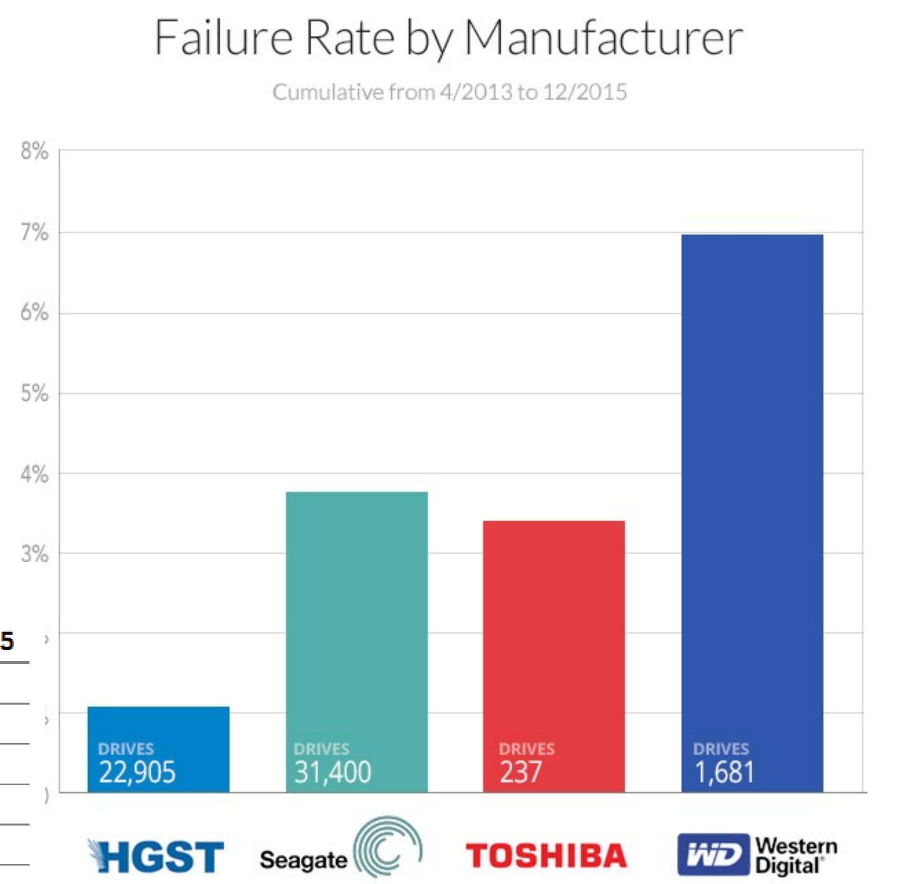
- Statistics for 56K disks of the Backblaze data center





Cumulative Failure Rate through the Period Ending

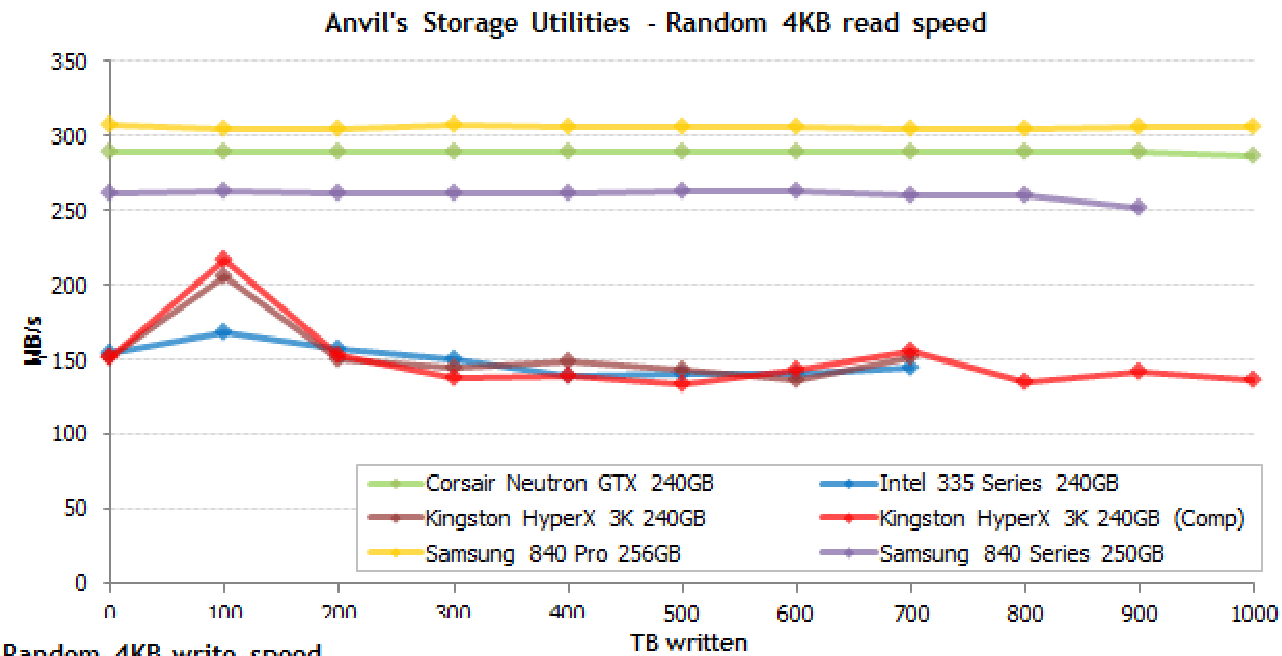
MFG	Model #	Highest QTY	12/31/13	12/31/14	12/31/15
HGST	HDS5C3030ALA630	4,596	0.9%	0.7%	0.8%
HGST	HDS723030ALA640	1,022	0.9%	1.8%	1.8%
Seagate	ST3000DM001	4,074	9.8%	28.3%	28.3%
Seagate	ST33000651AS	325	7.3%	5.6%	5.1%
Toshiba	DT01ACA300	58	-	4.8%	3.8%
WDC	WD30EFRX	1,105	3.2%	6.5%	7.3%

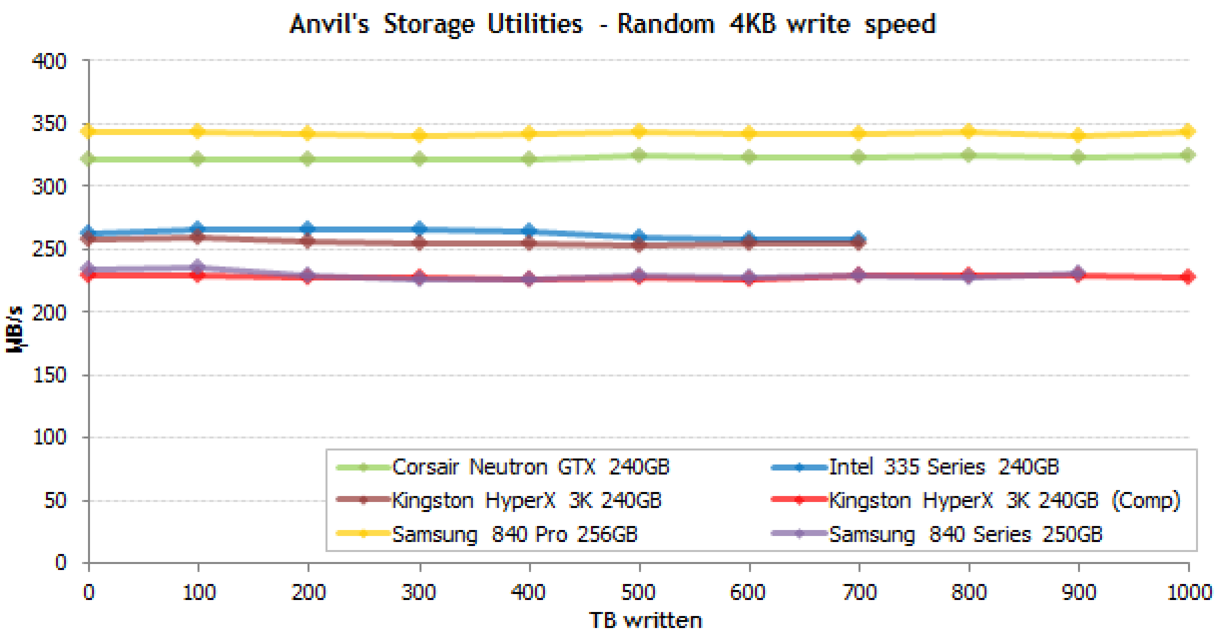


(HGST is the former Hitachi Global Storage Technologies)

BACKBLAZE

- SSD reliability





50GB / day
→ > 40 years lifetime

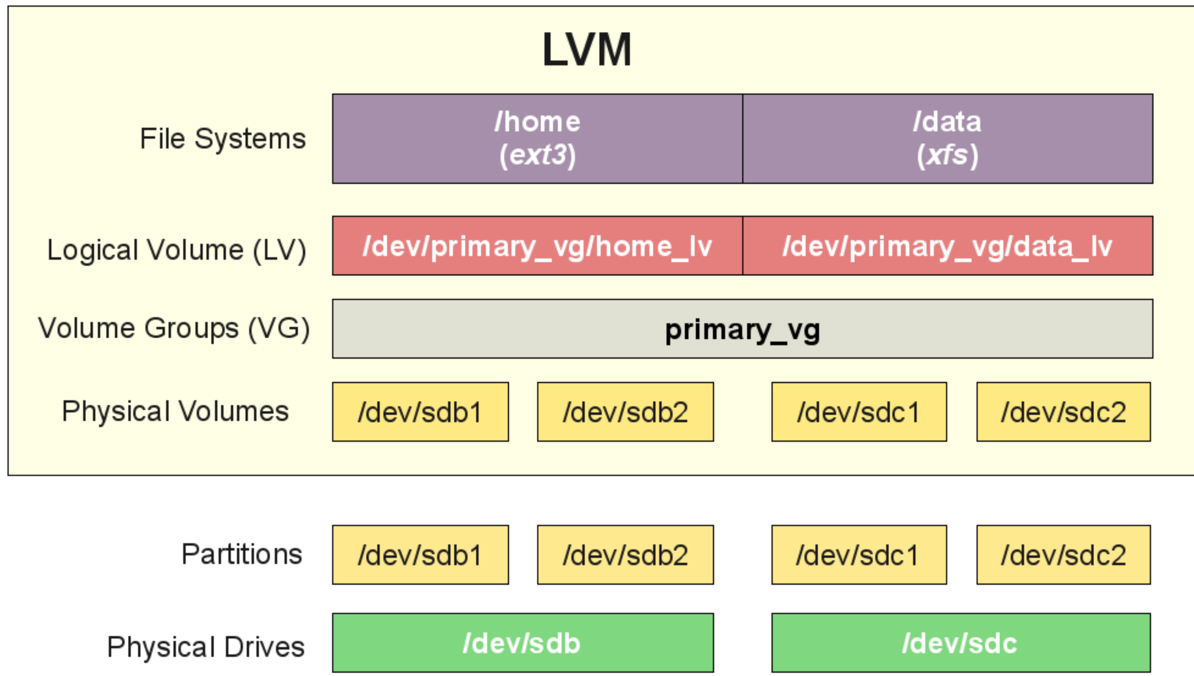
Source: http://techreport.com/review/24841/introducing-the-ssd-endurance-experiment

Virtual storage systems

- Overcoming the limits of physical storage solutions
  - capacity, performance, reliability
  - better management
  - better error recovery
  - unifying physical storage devices
- Virtual storage
  - implements a software storage layer that
    - is backed by other storage devices (physical or virtual)
    - unifies the underlying storage into a coherent system
  - examples: RAID, LVM (Linux), LDM (Windows)
  - they are also part of certain file system implementations (e.g. btrfs)

Virtual storage: Logical Volume Management

- An allocation system beyond the boundaries of the physical devices
  - more flexible management than partitions
  - logical volumes can be created from partitions, disks and other resources
  - e.g. Windows: Logical Disk Manager, Linux: Logical Volume Manager



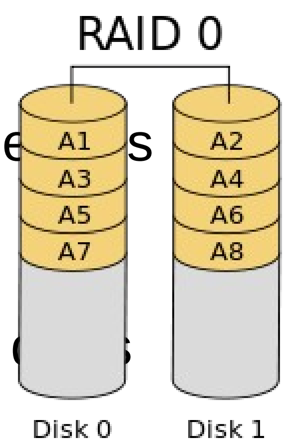
- Parts of the LVM
  - **Physical volumes**: disks, partitions, ...
  - **Logical volumes** virtual partitions
  - **Volume group**: a set of LVs – virtual storage
  - Allocation units
    - **Physical extents**: parts of the PV-s
    - **Logical extents**: LE-s are assigned to PE-s (1-N)
    - Usually N=1 ⇒ 1 logical unit is stored by 1 physical unit
    - RAID may use it differently (see later)

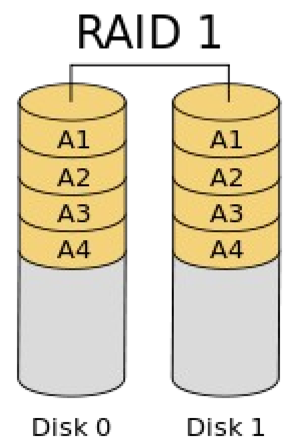
Virtual storage systems: RAID

- Redundant Array of Inexpensive Disks
  - „cheap” (smaller capacity) disks merged together
    - recently: “I” means Independent, RAID disks aren't cheap
  - goal: improve redundancy (reliability) and performance
  - HW and SW implementations
    - mainboard RAID implemented in software (cheap)
    - RAID boards: HW solution (expensive)

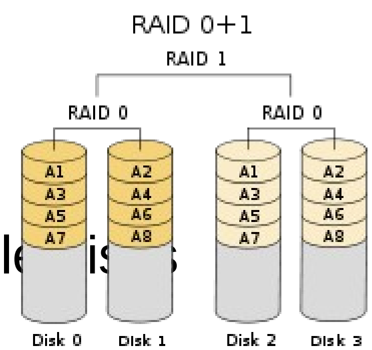
- Reliability
  - more disks mean more failures
    - 1 disk MTTF: 100 000 hours, 100 disk MTTF: 1000 hours (41 days)
    - how can we increase the reliability?
  - introducing redundancy
    - storing error correcting data
    - simple example: mirroring – storing the data twice
    - better: a parity bit can also detect a single error and correct it

RAID levels: 0 - 1

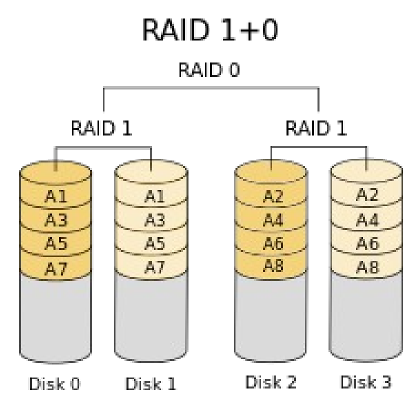




- RAID level: the mode of merging the physical devices
  - How the data is distributed on the N disks
- RAID 0 (stripe): the data is distributed on the N disks
  - Goal: improve performance
  - It can increase the throughput and the latency also
  - The disks capacities are combined
  - Failure of 1 disk ⇒ data loss



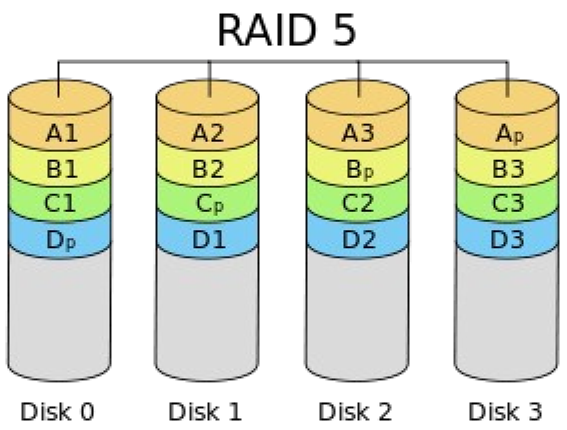
- RAID 1 (mirror): the same data are stored on multiple disks
  - Goal: improve reliability
  - The combined capacity is the size of a single disk
  - Slower write operations, read can be faster



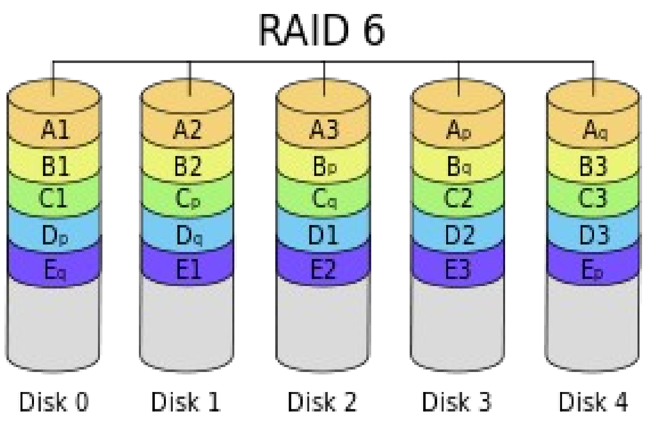
- Hybrid (nested) RAID solutions
  - RAID 01 (0+1): mirror of stripes
    - Rather a theoretical possibility, not used in practice
  - RAID 10 (1+0): stripe of mirrors
    - Great performance, improved reliability

Widely used RAID levels

- RAID 5
  - block-level striping with distributed parity
  - N+1 disk fault tolerance
  - a parity block is assigned to a group of data
  - this block is distributed among the disks
  - the performance is close to RAID0
  - the capacity is smaller with a size of 1 disk
  - Main problem: **silent error**



- RAID 6
  - block-level striping with double distributed parity
  - N+2 disk fault tolerance
  - extension of RAID5 with an additional parity block
  - no significant performance degradation
  - the capacity is smaller with a size of 2 disks
  - handles silent errors



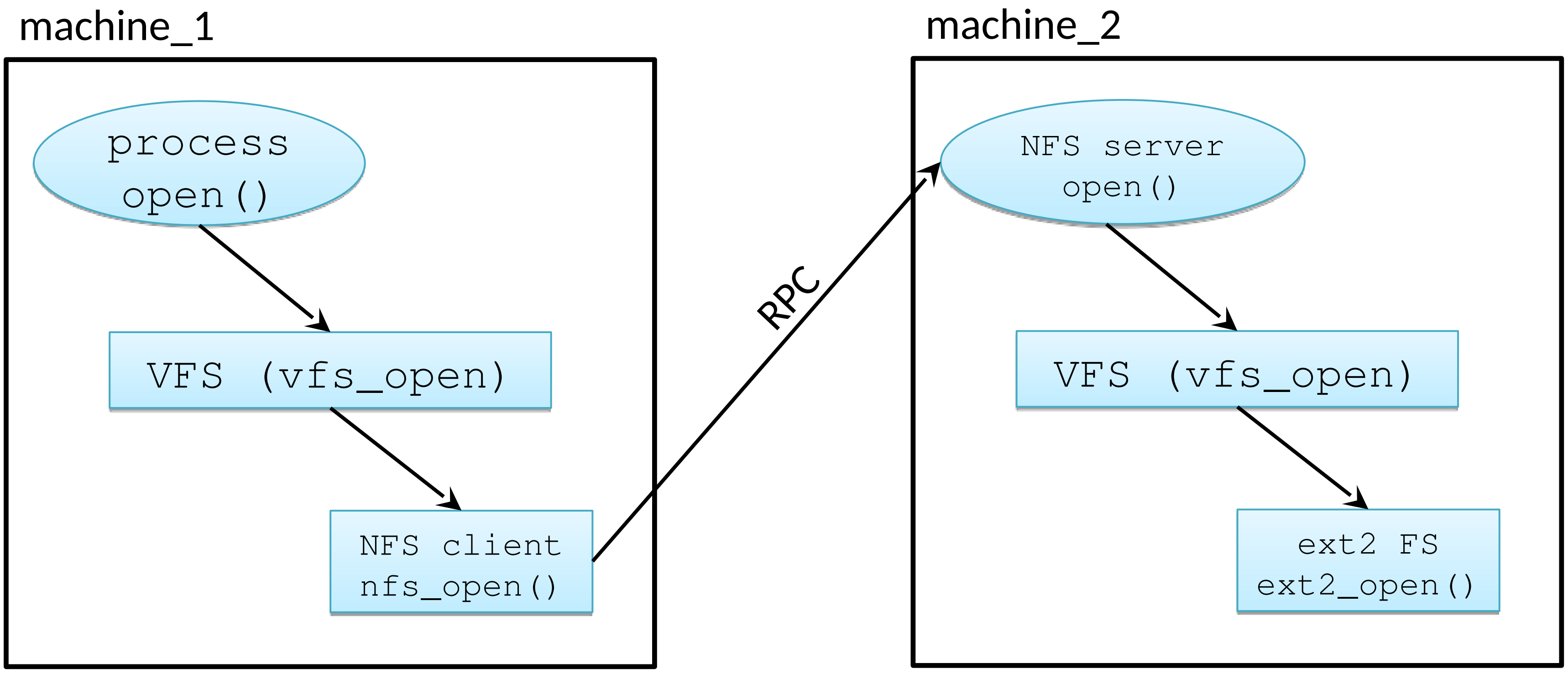
The limits of RAID systems

- How long does it take to correct an error (off-line)?
  - In the case of 4+1 disks (RAID5)
    - 150 GB disks: ~10 hours
    - 6 TB disks: ~80 hours
  - Spending days with error recovery is not acceptable
    - hot spares and RAID6 may improve the situation
- RAID needs the same type of disks
  - the replacement can be difficult
- RAID is a bonded structure, not flexible
  - cannot upgrade a RAID5 system to RAID6
  - transition takes a long time (off-line)
- Limited combined storage capacity
  - 6-8 disks at maximum
- RAID only protects against disk errors
  - What happens if the motherboard, CPU, RAM, power supply has an error?

Network and distributed file systems

- Goal: access to files stored in remote machines, sharing files
- Client-server based storage systems
  - Server: provides access to the local storage system
  - Client: connects to the server and grants access to the remote data
  - **Network Attached Storage (NAS)**
    - High-level, file oriented transmission
    - NFS (Network File System), see next slide
    - SMB/CIFS (Common Internet File System) – Network file system of Windows
  - Block level network storage: **SAN (Storage Area Network)**
    - Low level data transmission
    - iSCSI (internet SCSI): for transmitting SCSI commands over IP
- Distributed file system
  - Operates as a distributed system
  - The data storage is distributed amongst the nodes of the system
  - Examples:
    - Ceph (RedHat, SUSE), Google GFS, RedHat GlusterFS,
    - Windows DFS, PVFS, Orange FS
- Challenges: latency, network errors, consistency

A simple implementation of NFS using VFS and RPC



Challenges of network file systems

- Location: where is the data stored?
  - Location transparency
    - The name/path of the files are not referring the location
  - Location independency
    - The names and paths don’t change when the data is moved
- Question of network copies
  - The requests are served by remote services
    - Every operation should be performed on a single instance of the data
    - The network introduce latency and possible errors
    - The order of the operations are critical
  - The requests are served with the help of temporary local storages
    - the local machine maintain a copy of the data
    - Size is limited by the local machine
    - Multiple instances ⇒ consistency problems
- Operation of the network server
  - stateful: the file operations have a state (faster)
  - stateless: slower, but more reliable

Scalable, distributed storage systems: Ceph

- Universal, virtual storage systems (SW implementation)
  - Block based system (SAN)
  - File based system (NAS)
  - Object store (OSD)
- Scalable, fault tolerant
  - no single point of failure
  - Every component is replaceable at runtime (disc, machine)
  - Dynamic configuration (level of replication)
- Further advantages
  - PB capacity
  - Significantly faster error recovery than RAID
  - No special HW
  - Hot spares are not required (see RAID spare disk)
  - Cooperates with other virtualization systems (OpenStack, Amazon S3)
  - Open source

Further development of storage systems

- Integrated file and storage systems
  - Integrating the file systems with the solutions of RAID and LVM
  - e.g. zfs, btrfs
- Scalability
  - dynamic change of storage capacity (runtime)
- Reliability
  - large capacity → many disks → high possibility of errors
  - The error correction time should be eliminated
- Memory based storages
  - The SSD’s speed is reaching the speed of the physical memory → new principles of development
- Data deduplication (e.g. zfs, btrfs)

- Further reading
  - Microsoft ReFS (Resilient File System)
  - Solaris ZFS (Z File System)
  - Linux Btrfs (B-Tree File System, „butter F S”)
  - F2FS (Flash-Friendly File System, Samsung)
  - GPUfs (file access on GPU-s, see heterogenous multiprocessor systems)